

Q.1)

```
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Ramesh   | CSE       |
| Suresh   | CSE       |
| Mahesh   | ECE       |
| Naresh   | IT        |
+-----+-----+
4 rows in set (0.00 sec)
```

Q.2)

```
+-----+-----+-----+
| name   | email           | role_name |
+-----+-----+-----+
| Ajay   | ajay@mail.com   | Admin     |
| Vijay  | vijay@mail.com  | NULL      |
| Ramesh | ramesh@mail.com | Manager   |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Q.3)

```
+-----+-----+
| name   | role_name |
+-----+-----+
| Ajay   | Admin     |
| Ramesh | Manager   |
| NULL   | Developer |
+-----+-----+
3 rows in set (0.00 sec)
```

Q.4)

```

--> FROM users u RIGHT JOIN roles r ON u
+-----+-----+-----+
| name   | email           | role_name |
+-----+-----+-----+
| Ajay   | ajay@mail.com   | Admin     |
| Vijay  | vijay@mail.com  | NULL      |
| Ramesh | ramesh@mail.com | Manager   |
| NULL   | NULL            | Developer |
+-----+-----+-----+
4 rows in set (0.00 sec)

```

Q.5)

```

+-----+-----+
| Employee | Manager |
+-----+-----+
| Ramesh   | NULL    |
| Suresh   | Ramesh  |
| Mahesh   | NULL    |
| Naresh   | Suresh  |
+-----+-----+
4 rows in set (0.01 sec)

```

Q.6)

```

+-----+-----+-----+-----+
| order_id | customer_name | city      | total_amount |
+-----+-----+-----+-----+
| 401      | Anil          | Delhi     | 5000.00      |
| 402      | Anil          | Delhi     | 10000.00     |
| 403      | Ravi          | Hyderabad | 8000.00      |
| 404      | Sunil         | Mumbai    | 6000.00      |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

```

Q.7)

```

--> GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | EmpCount | AvgSalary | MaxSalary |
+-----+-----+-----+-----+
| CSE       | 2        | 67500.000000 | 75000.00 |
| ECE       | 1        | 55000.000000 | 55000.00 |
| IT        | 1        | 45000.000000 | 45000.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

Q.8)

```
WHERE salary > 50000 AND dept_name = 'CSE'
```

emp_name	salary	dept_name
Ramesh	75000.00	CSE
Suresh	60000.00	CSE

2 rows in set (0.00 sec)

Q.9)

customer_name	TotalPurchase
Anil	15000.00
Ravi	8000.00
Sunil	6000.00

3 rows in set (0.00 sec)

Q.10)

student_name	course_name	faculty_name
Jay	DBMS	Prof. Rao
Kiran	OS	Prof. Mehta

2 rows in set (0.00 sec)

Q.11,12,13,14,15)

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance=balance-5000 WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE accounts SET balance=balance+5000 WHERE account_no=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance=balance-3000 WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE accounts SET balance=balance+3000 WHERE account_no=999;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> UPDATE accounts SET balance=balance+2000 WHERE account_no=101; -- Ajay = 47000  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> SAVEPOINT s1;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> UPDATE accounts SET balance=balance-1000 WHERE account_no=101; -- Ajay = 46000  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> ROLLBACK TO s1; -- undo -1000, back to 47000  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> COMMIT;  
Query OK, 0 rows affected (0.01 sec)
```